EXPERIMENT NO: 6

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Subject Name: System Design
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Aim

To design a highly scalable video streaming platform similar to YouTube that supports adaptive bitrate streaming (HLS/DASH), dynamic manifest handling and real-time engagement features (likes and comments) with high availability and low latency.

Objectives

• Understand Adaptive Bitrate Streaming using HLS and DASH protocols.
• Implement scalable like and comment systems videos.
• Ensure fault tolerance, high availability, and eventual consistency.

Functional Requirements

1. Users can sign up and log in using their emails.
2. Users can upload videos.
3. Users can watch videos.
4. Users can like/dislike videos.
5. Users can comment on videos.
6. Users can search videos.
7. System should support multiple resolutions (360p, 720p, 1080p).

Non-Functional Requirements

1. Scalability: 100M+ DAU

2. CAP Theorem: Availability >>>> Consistency

Users should always be able to watch videos
Target: 99.99% uptime

3. Latency: 50-80 ms
User should be able to see the videos with zero or negligible buffering

Core Entities of the System

1. User

2. Video
3. Likes and Comments
4. Channel

API Design

User-Onboarding

1. POST Call /api/v1/auth/signup
2. POST /api/v1/auth/login

Upload a Video

POST Call /videos/upload

Request {
title,
description, video_file
}

Get Video

GET Call /videos/{video_id}

Returns {
title
description
views video_url
}

Like & Dislike video

1. POST Call /api/v1/videos/{video_id}/like
2. POST Call /api/v1/videos/{video_id}/dislike

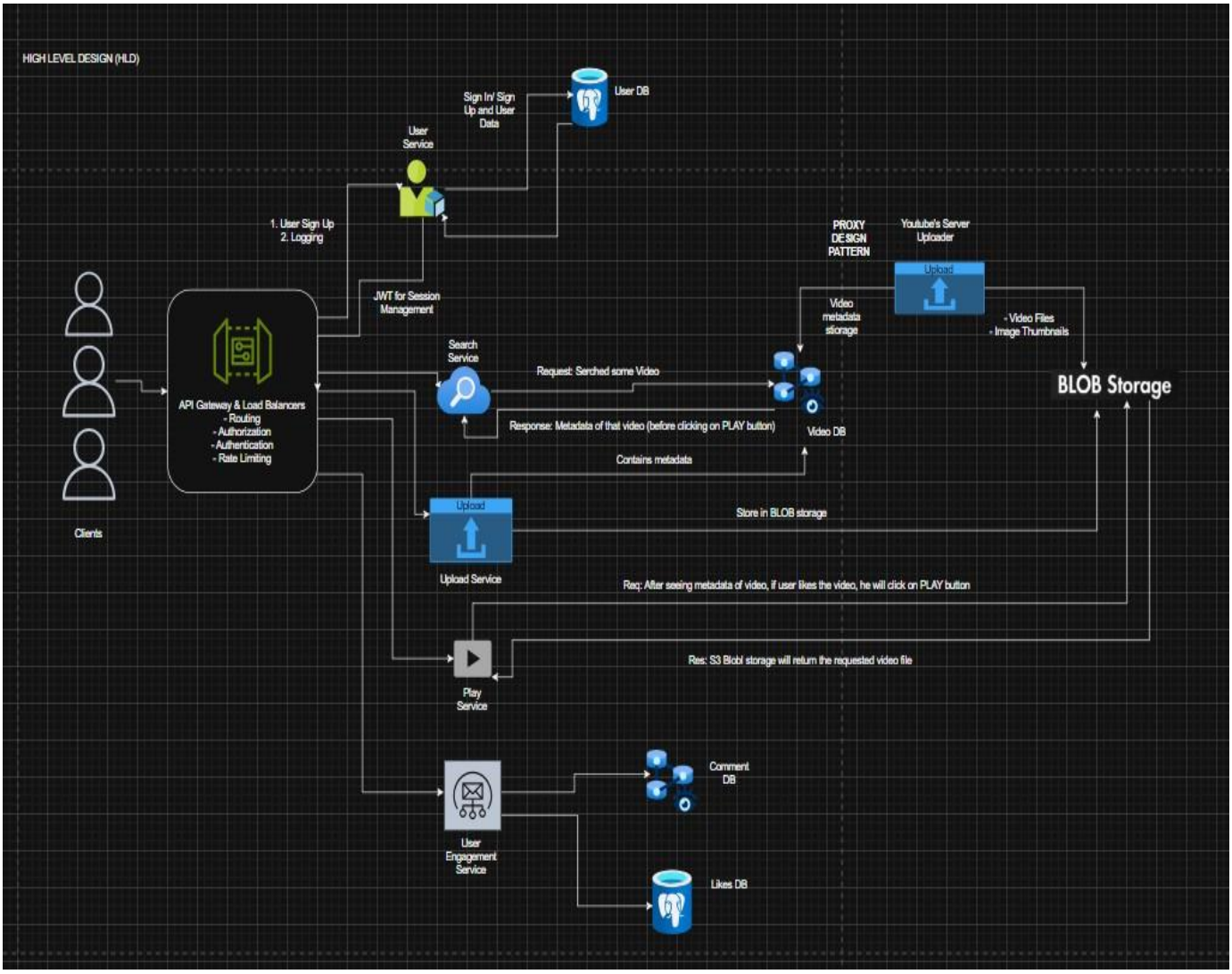
Comment on Video

POST Call /videos/{video_id}/comment

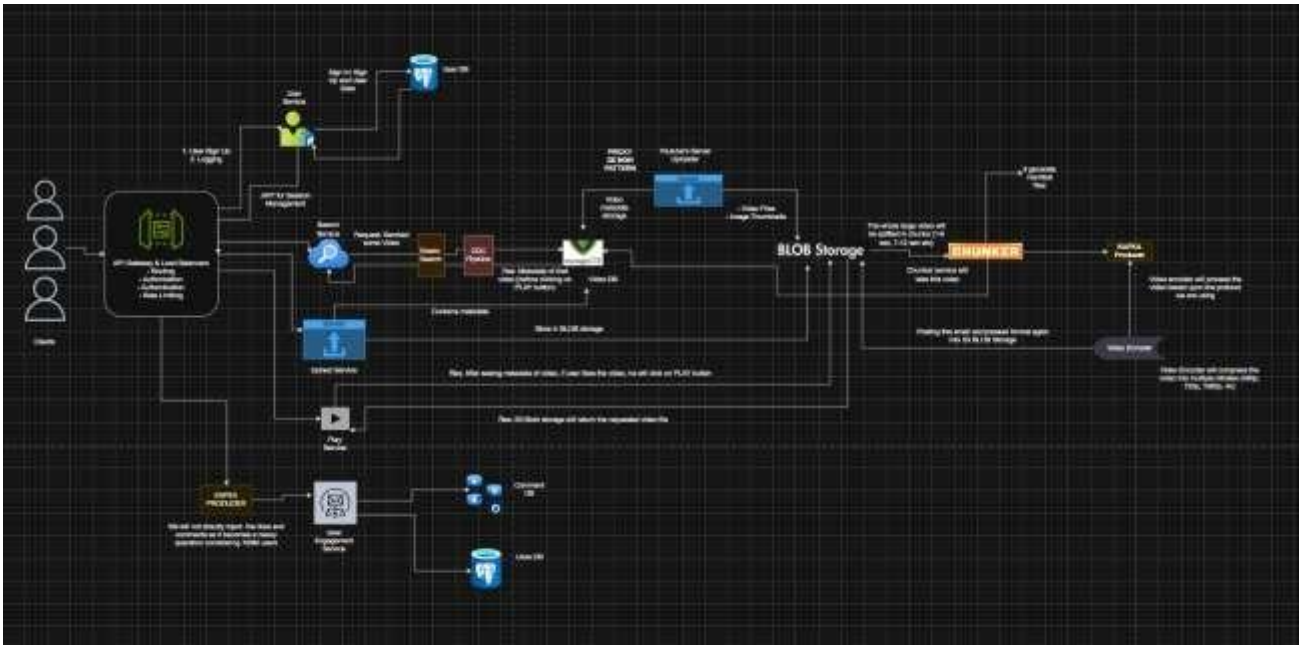
Search Video

GET Call /videos/search?q=music

High Level Design (HLD)



LOW-LEVEL DESIGN (LLD)



Outcome

- Understand the architecture of a scalable video streaming platform similar to YouTube.
- Learn how adaptive bitrate streaming using HLS and DASH improves video playback under varying network conditions.
- Design scalable APIs and data models to support real-time engagement features such as likes and comments.
- Apply distributed system principles (scalability, high availability, low latency, and eventual consistency) in large-scale applications.